Mitigating Bias in Vision-Language Models

Author: Claudia Cornacchia
Course: Computer Vision
Date: 22/01/2026

Sapienza University of Rome

Outline

- Problem Statement
- State of the Art
- Proposed Methods
- Datasets
- Model evaluation strategy
- Results
- Conclusions
- References

Problem Statement

- **Vision-Language Models** (VLMs) such as **CLIP** achieve strong zero-shot generalization thanks to large-scale pre-training on uncurated web data.

- Problem: this training paradigm causes models to inherit **societal stereotypes** (in gender and ethnicity) present in the source data, raising ethical concerns for real-world applications where model's outputs can influence decision making.

Project objective: develop a technique to mitigate **gender and ethnicity** bias in CLIP for a set of **predefined professions**, while maintaining the zero-shot performance on domain-specific datasets.

State of the Art

How current research approaches bias in VLMs?

Data balancing

- Constructing dataset where demographic groups are equally represented.
  - *CLIP THE BIAS* [1] reweighting algorithm to balance gender proportion and remove correlations between gender and occupation.

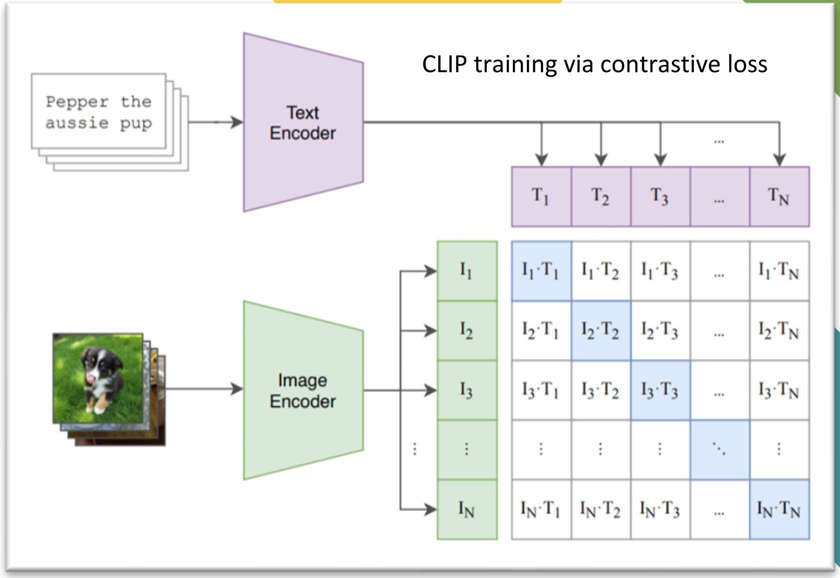
Fine-tuning

- *CLIP THE BIAS* [1] train and fine-tune CLIP using the balanced dataset.
- *LightFair* [10] **LoRA** on the text encoder to make neutral job prompts equally distant from all demographic groups.

Inference-time Ablation

- Remove bias during inference. Identify the bias direction in the embedding space and project the inputs onto the null space of those directions, making the model blind to the protected attributes [3] [4].
  - *Tanjim et al.* [4] compute the bias subspace using **PCA.**

Proposed Method (1/2): Fine-tuning on balanced dataset




```
trainable params: 491,520 || all params: 151,768,833 || trainable%: 0.3239
```

- **LoRA**: freezes original weights and trains only adapter layers, obtaining a 99% reduction in trainable parameters.
- **Contrastive Loss:** using **neutral prompts** and the balanced **DebiasingDB** (same number of samples for every demographic group within each profession), ensures that the model sees every neutral profession prompt paired with each demographic group equally often, forcing it to unlearn specific profession-group associations.
- Baseline: **CLIP ViT-B/32** (smallest and easiest to fine-tune).
- **Validation set**: prevent the model from overfitting to the synthetic fine-tuning dataset

<u>Two fine-tuning strategies:</u>

- **Vision only**: hypothesis that bias is primarily driven by the vision encoder over-attending to demographic traits
- **Joint vision-text**: social stereotypes are also encoded in the text embeddings where profession words are semantically closer to specific demographic groups

Proposed Method (2/2): Inference-time Ablation

A training free strategy that handles intersectional bias using a subspace-based projection method [4][5][6].

1. **Demographic deviation vector**, that isolates the demographic information encoded in CLIP's embedding space.

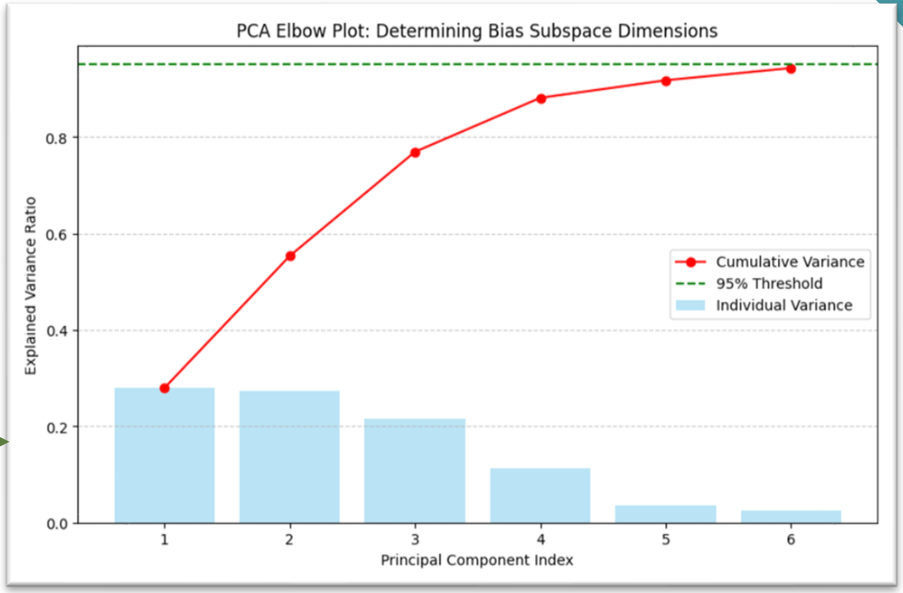
$$\overrightarrow{d_G} = \mu_G - \mu_{global}$$

2. **Bias subspace** estimation U_{bias}: I stacked the vectors together and applied **PCA** to obtain the orthogonal directions in the embedding space that maximize demographic variance.

3. **Low-rank** bias subspace: 6 components (k) for images and 7 for text explain more than 94% of the demographic variance.

4. **Final projection**: during inference project the input embedding (v) onto the null space of the bias subspace.

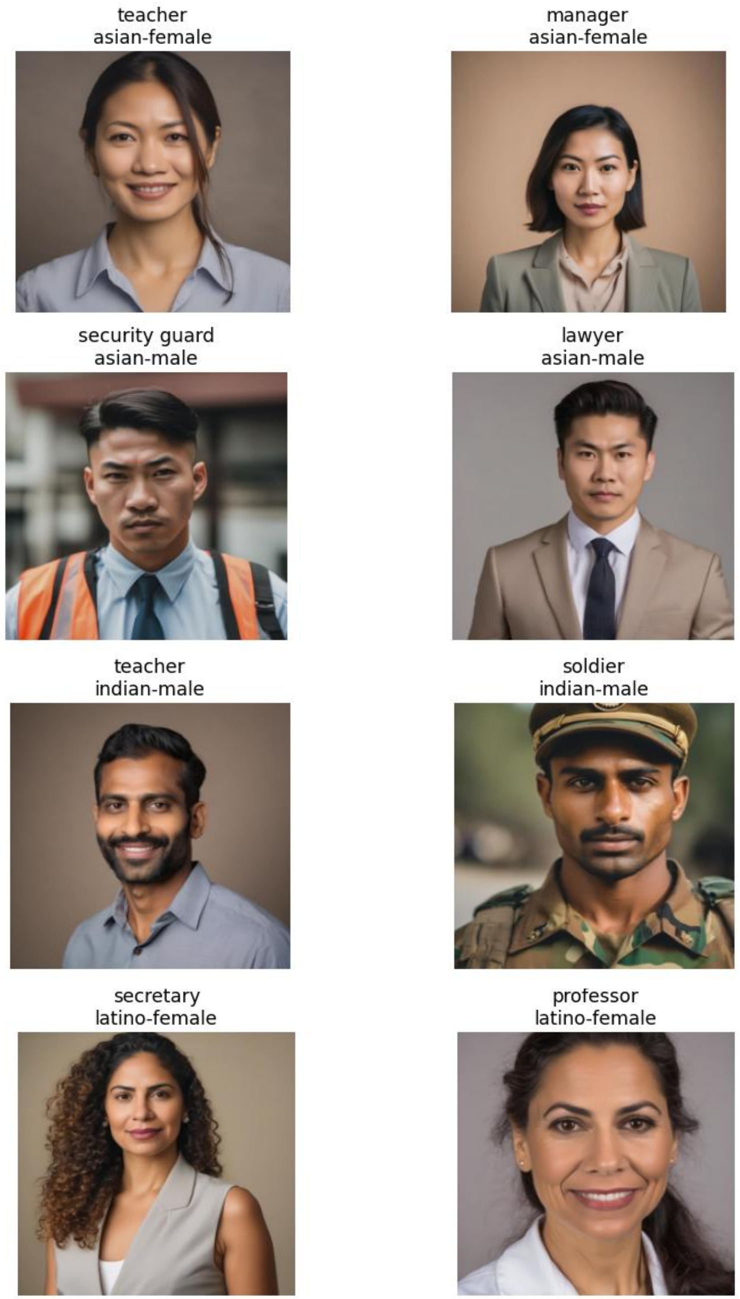
$$v_{debiased} = v - \sum_{i=1}^{k} (v \cdot u_i) u_i$$



Dataset (1/2)

1. Debiasing dataset:

- Synthesized by aggregating 12 datasets from the **NYUAD-ComNets** AI generated images collection. Is perfectly balanced across the intersection of **2 genders and 6 ethnicities**, ensuring that for every profession the model sees every demographic group with equal probability.
- Designed to address the two main forms of dataset bias [1]
- Divided in **Fine-tuning set** (12k images), and **Validation set** (3k images).



Dataset (2/2)

2. Debiasing evaluation dataset:

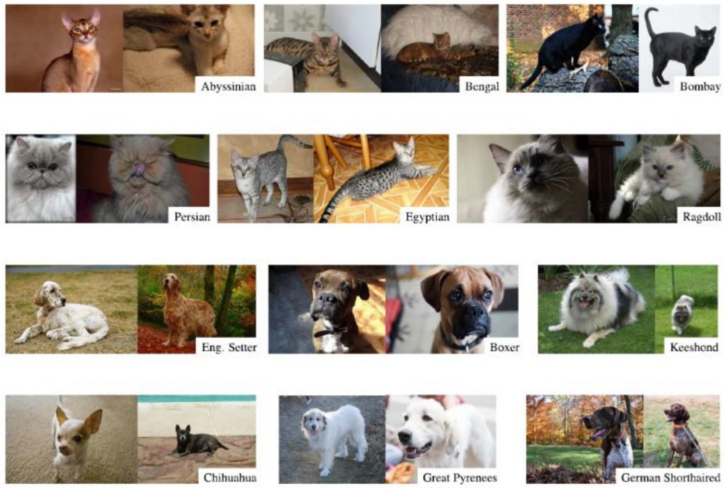
- **FairFace**, 100k real-world face images across 7 ethnicity groups.
- I filtered out children with age between 0-9.

FairFace

3. Zero-shot evaluation dataset:

- **Oxford-IIIT Pet** (37 categories, 200 images per class)
- **Food-101** (large scale dataset of 101 food categories)

Food 101



Oxford-IIIT Pet

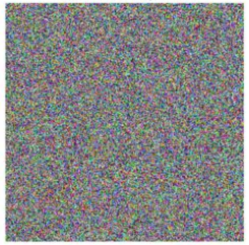
Model evaluation strategy

1. Representation bias:

- Given in input a Gaussian noise image (no visual information) a fair model should assign equal probability to all (m) demographic groups.
- RB: measures the maximum deviation from uniform distribution.

$$RB = \max_{k \in [m]} \left| E_D[f_k(x)] - \frac{1}{m} \right|$$

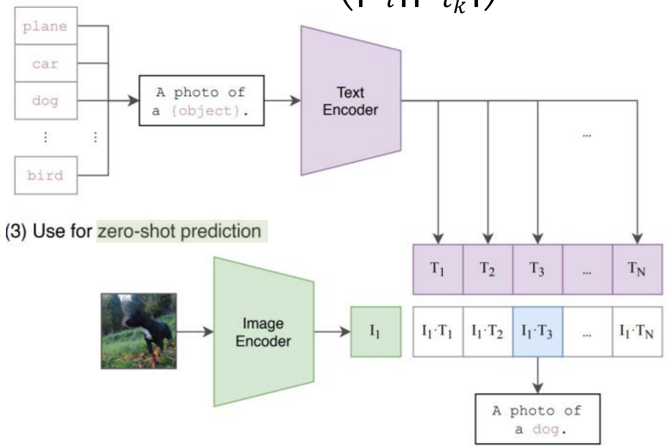
Noise sample 1



2. Zero-Shot generalization:

- Compute cosine similarity between the image and text embeddings for all (k) possible class prompts.
- Measure **Top-1 Accuracy**, number of predictions ($\hat{y}$ prompt with highest similarity score) that matches the true class.

$$\hat{y} = \text{argmax}_k \left(\frac{e_i \cdot e_{t_k}}{|e_i||e_{t_k}|} \right)$$



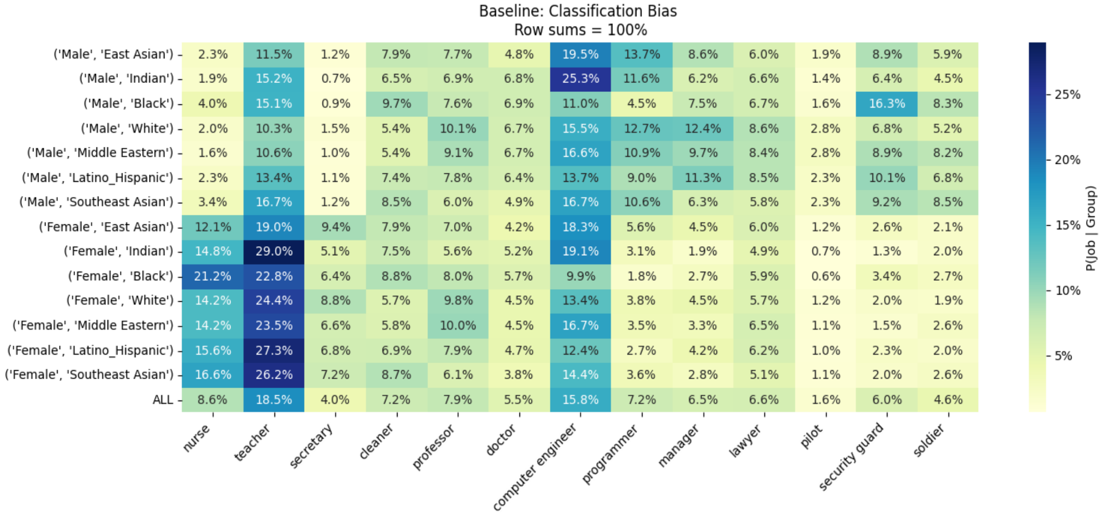
3. Occupation bias:

- FairFace has gender + ethnicity labels, not profession, so instead of accuracy I analyze CLIP's predicted profession distributions (via cosine similarity scores) to find stereotypical associations between demographic groups and jobs.
- **Classification**: P (Profession|Group), measures **stereotyping**. Metric: **JSD**, how much a group profession distribution differs from global average.
- **Retrieval**: P(Group|Profession), measures **exclusion**. Metric: **MAD**, how far each group retrieval probability is from perfect equality

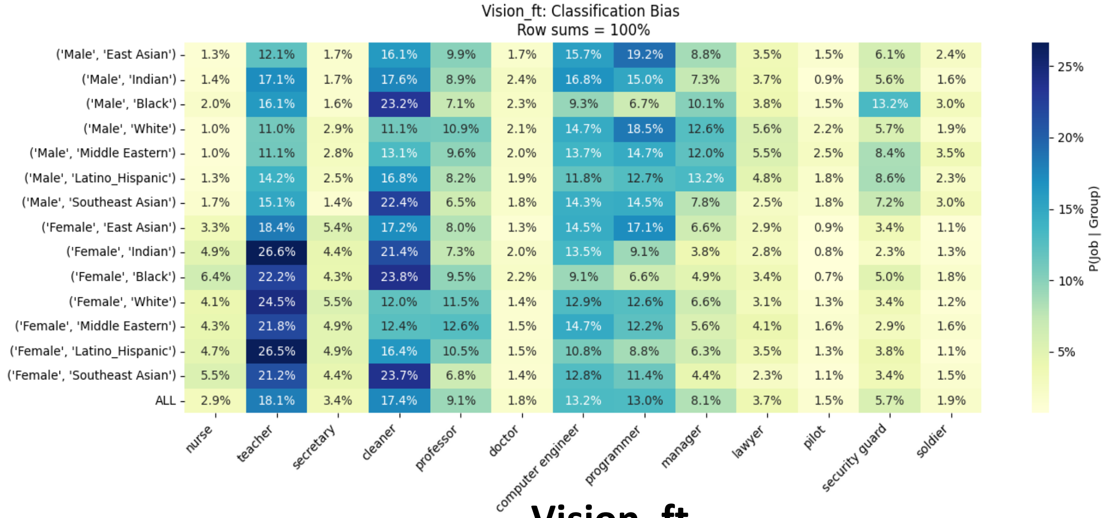
Vision-only fine-tuning: reduced classification bias, but it remains the language bottleneck.

Results

Model	Acc Oxford-IIIT Pet ↑	Acc Food-101 ↑	RB Score ↓	Mean JSD ↓	MAD ↓
Baseline	0.843	0.786	0.269	0.039	0.021
Vision_ft	0.805	0.803	0.289	**0.017**	0.014
Vision_text_ft	0.823	0.800	0.247	0.024	0.011
Ablated	0.819	0.815	0.032	0.002	0.004



Baseline

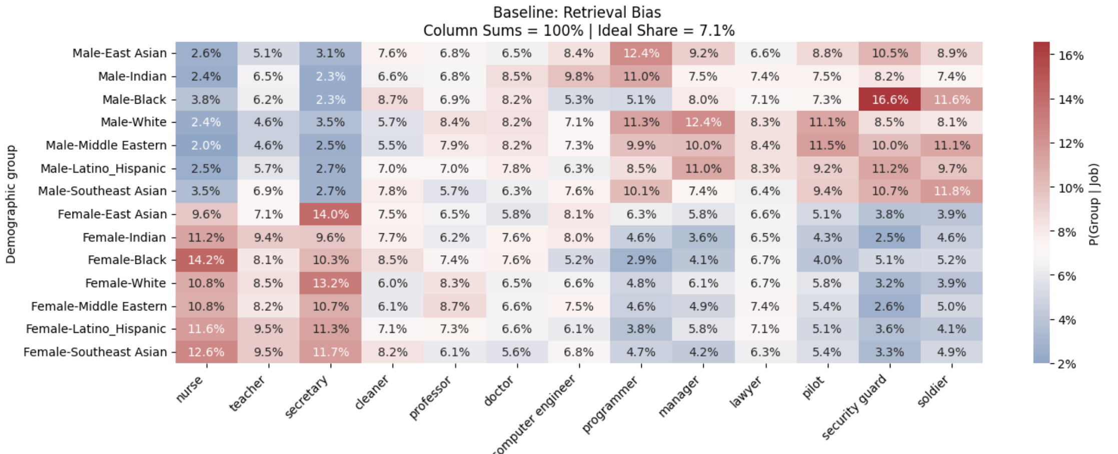


Vision_ft

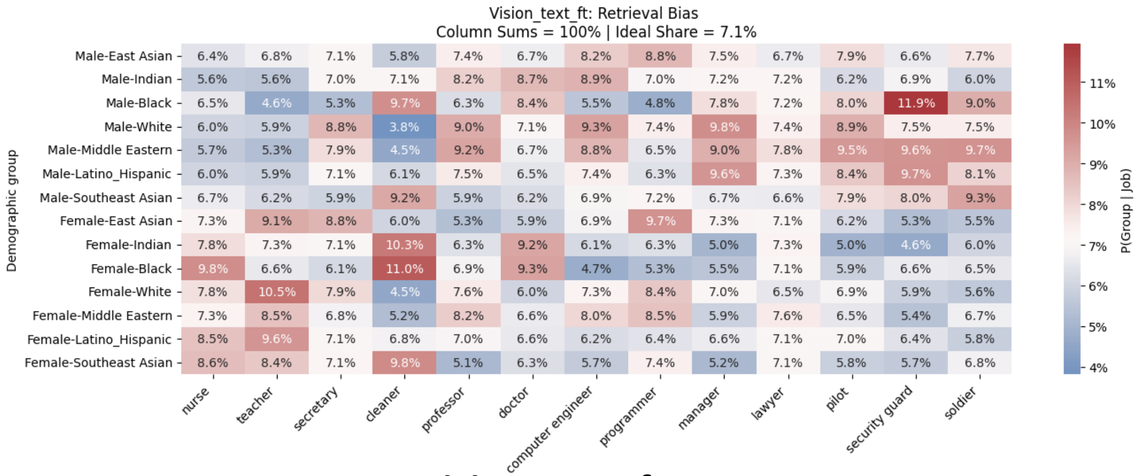
Joint vision-text fine-tuning: improved retrieval fairness but caused overfitting. The model learned shortcuts by assigning to minorities a single profession.

Results

Model	Acc Oxford-IIIT Pet ↑	Acc Food-101 ↑	RB Score ↓	Mean JSD ↓	MAD ↓
Baseline	0.843	0.786	0.269	0.039	0.021
Vision_ft	0.805	0.803	0.289	0.017	0.014
Vision_text_ft	0.823	0.800	0.247	0.024	**0.011**
Ablated	0.819	0.815	0.032	0.002	0.004



Baseline

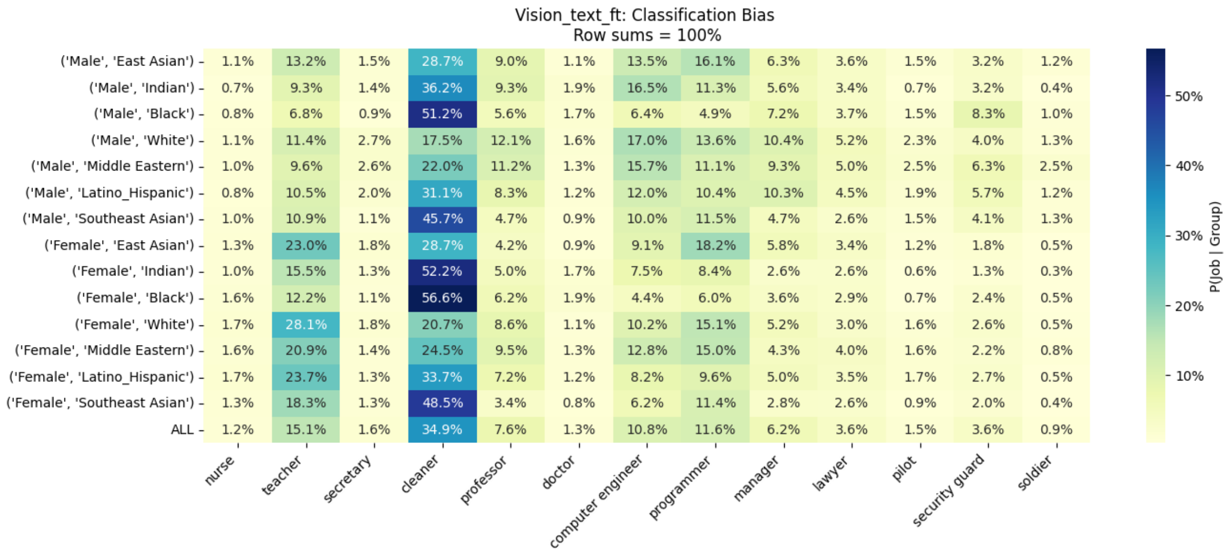


Vision_text_ft

Joint vision-text fine-tuning: improved retrieval fairness but caused overfitting. The model learned shortcuts by assigning to minorities a single profession.

Results

Model	Acc Oxford-IIIT Pet ↑	Acc Food-101 ↑	RB Score ↓	Mean JSD ↓	MAD ↓
Baseline	0.843	0.786	0.269	0.039	0.021
Vision_ft	0.805	0.803	0.289	0.017	0.014
Vision_text_ft	0.823	0.800	0.247	**0.024**	0.011
Ablated	0.819	0.815	0.032	0.002	0.004



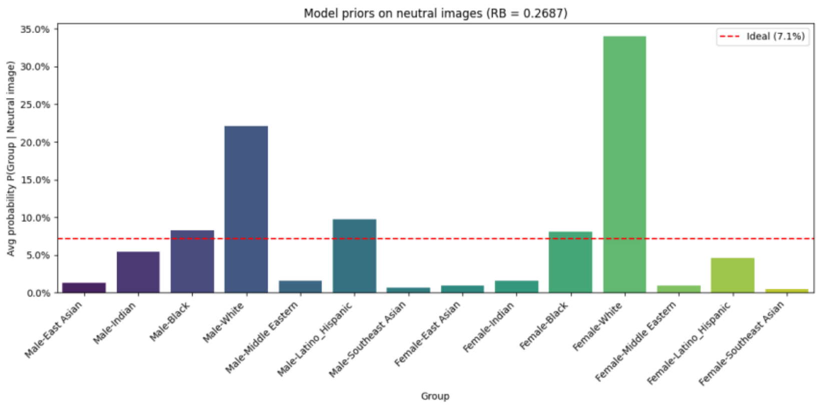
Vision_text_ft → overfitting

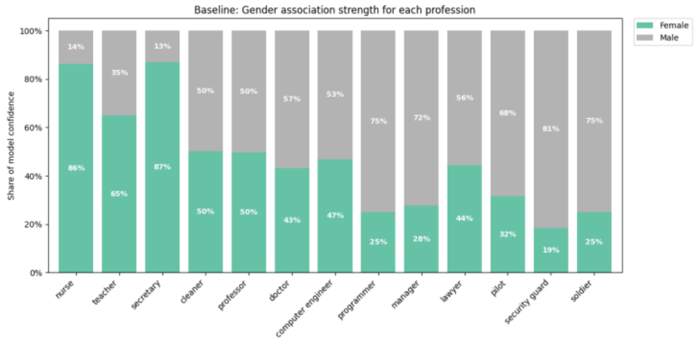
Results

Inference-time ablation: best results, generalizing well also to new dataset [3]

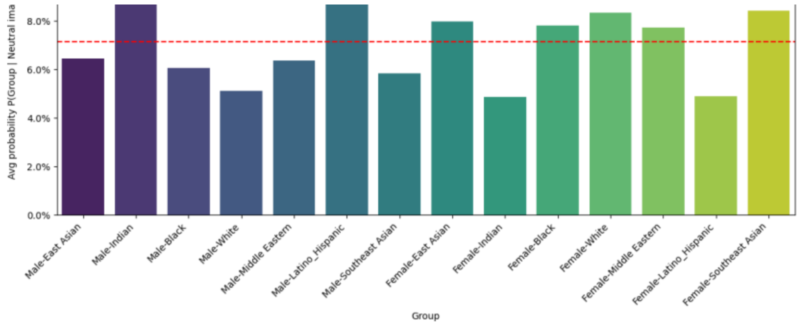
Model	Acc Oxford-IIIT Pet ↑	Acc Food-101 ↑	RB Score ↓	Mean JSD ↓	MAD ↓
Baseline	0.843	0.786	0.269	0.039	0.021
Vision_ft	0.805	0.803	0.289	0.017	0.014
Vision_text_ft	0.823	0.800	0.247	0.024	0.011
Ablated	0.819	0.815	**0.032**	**0.002**	0.004

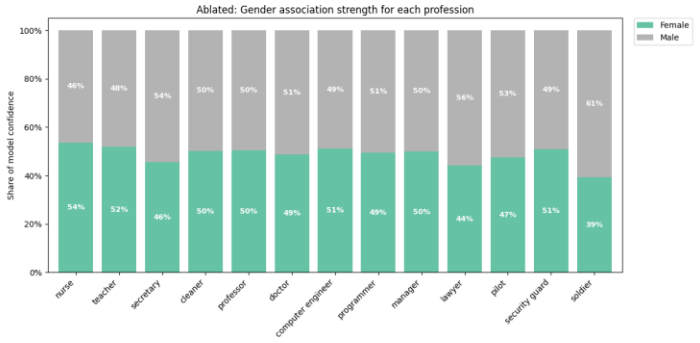
Baseline →





Ablated →



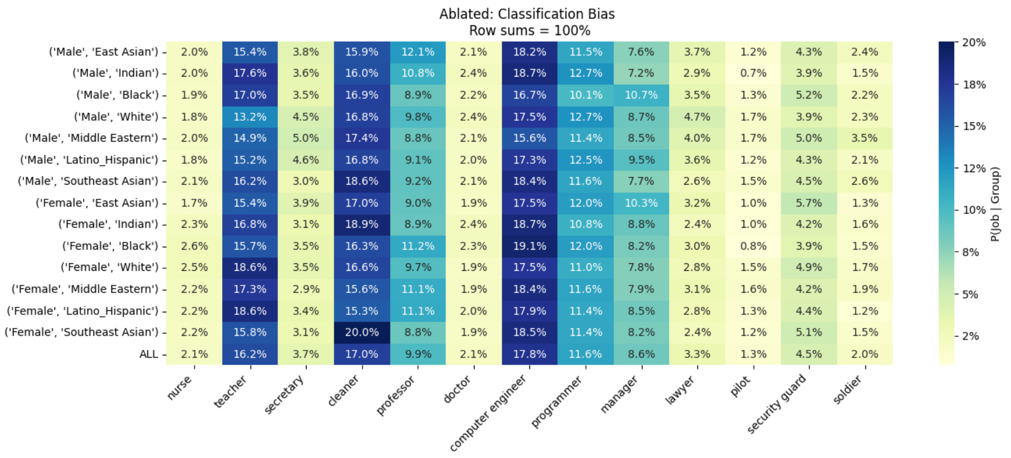


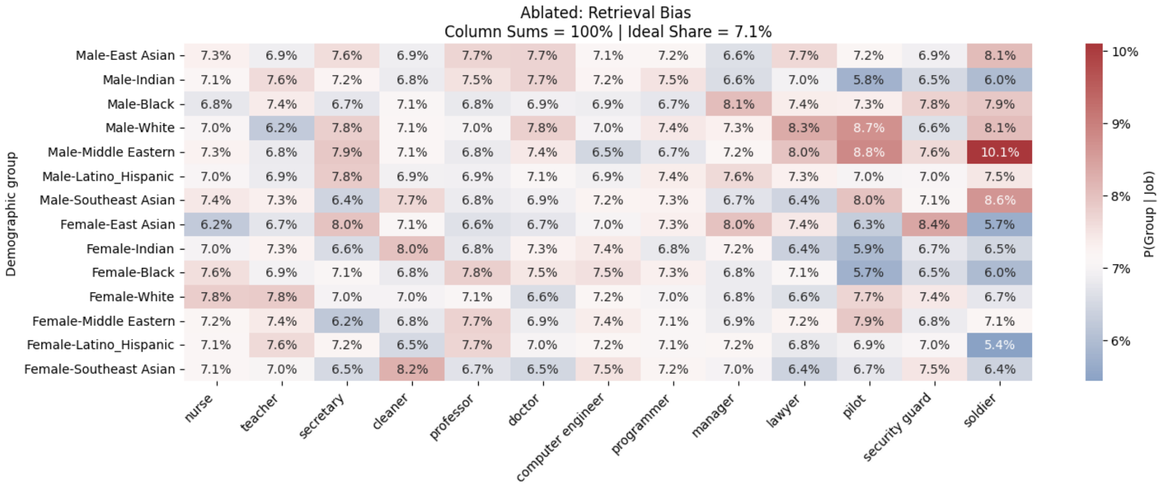
Results

Inference-time ablation: best results, generalizing well also to new dataset [3]

Model	Acc Oxford-IIIT Pet ↑	Acc Food-101 ↑	RB Score ↓	Mean JSD ↓	MAD ↓
Baseline	0.843	0.786	0.269	0.039	0.021
Vision_ft	0.805	0.803	0.289	0.017	0.014
Vision_text_ft	0.823	0.800	0.247	0.024	0.011
Ablated	0.819	0.815	0.032	**0.002**	**0.004**

Ablated →





Ablation Study

Different CLIP models, with similar architectures but trained on different datasets, exhibit very different bias behaviors

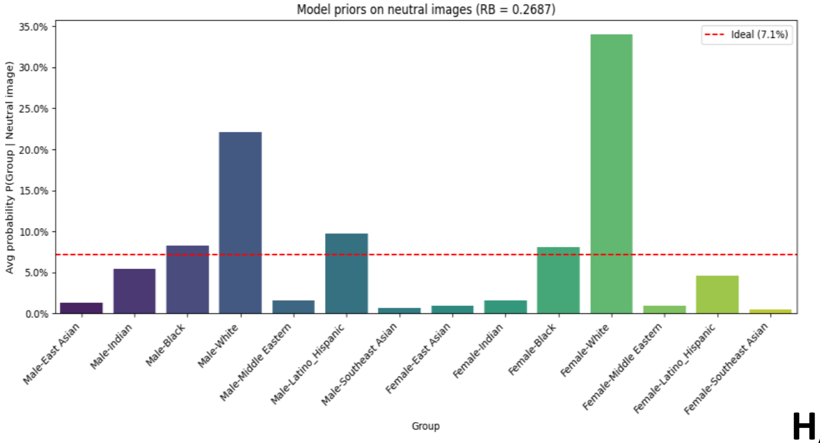
	Model	Acc Oxford-IIIT Pet ↑	Acc Food-101 ↑	RB Score ↓	Mean JSD ↓	MAD ↓
Baseline →	CLIP ViT-B/32	0.843	0.786	0.269	0.039	0.021
Architecture scale →	CLIP ViT-L/14	0.917	0.907	0.313	**0.029**	**0.017**
Using large scale web data →	CLIP-ViT-H/14-LAION-2B	0.919	0.908	**0.134**	0.059	0.102

- **Scaling** the architecture from ViT-B/32 to ViT-L/14 slightly **improve classification and retrieval** bias but **increases representation bias** with dominant preference shifts from *female–white* to *male-white*.
- Using **large scale web data** (CLIP-ViT-H/14-LAION-2B) achieves the **lowest representation bias**, probably due to the higher diversity of the LAION-2B dataset. However, it shows a **higher stereotyping and exclusion.**

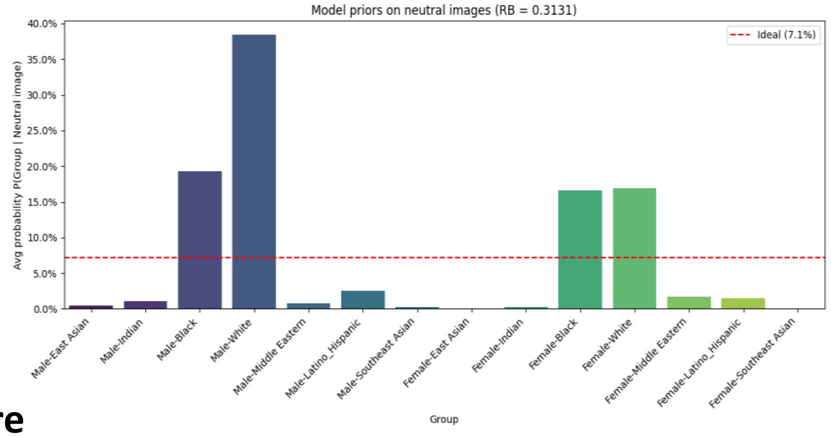
Ablation Study

Different CLIP models, with similar architectures but trained on different datasets, exhibit very different bias behaviors

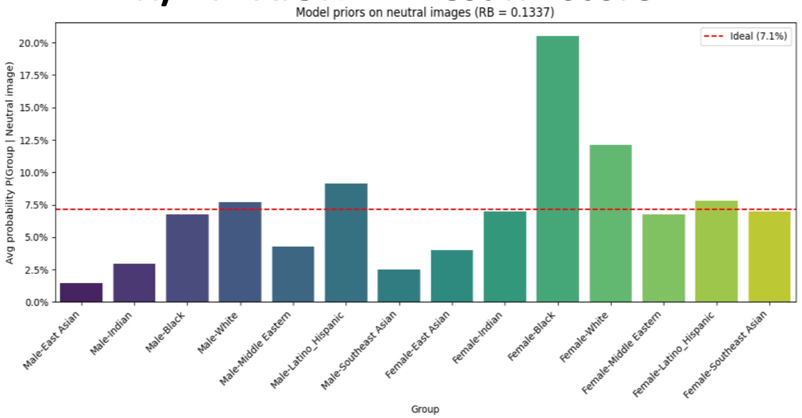
Baseline RB score



L/14 Worst RB score and preference shift



H/14-LAION-2B Best RB score

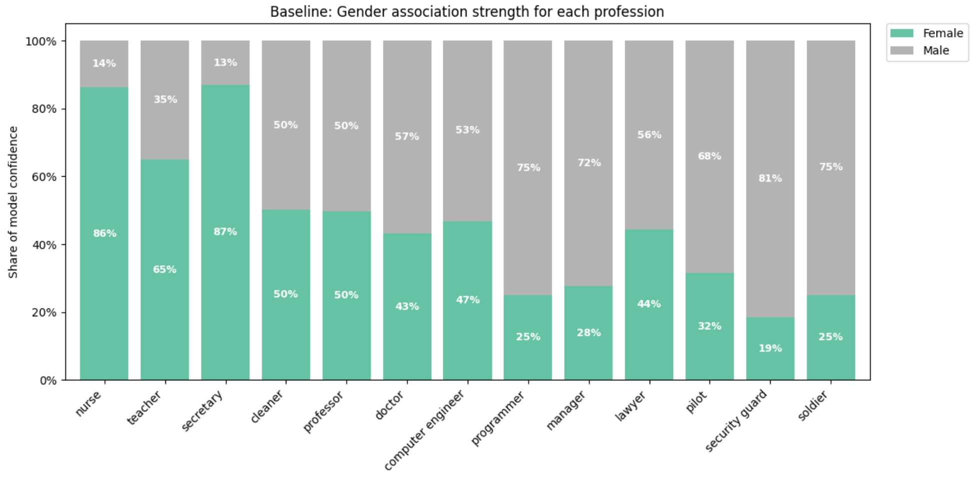


Ablation Study

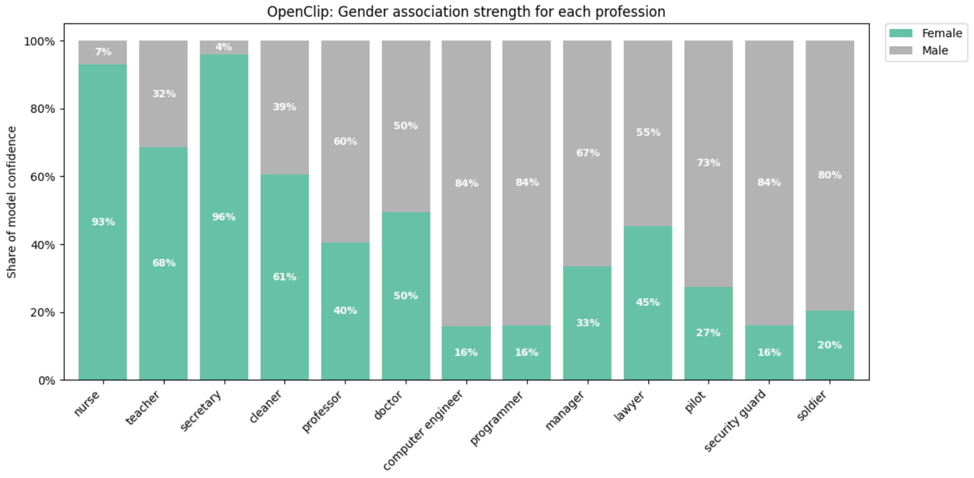
Different CLIP models, with similar architectures but trained on different datasets, exhibit very different bias behaviors

Stereotyping:

Baseline (CLIP ViT-B/32)



Using large scale web data (CLIP-ViT-H/14-LAION-2B)



Conclusion

- **Fine-tuning risks:** while data balancing and fine-tuning mitigates bias, this techniques are prone to overfitting, learning demographic shortcuts instead of true fairness.

- **Ablation superiority:** Inference-time ablation is the most robust strategy. It removes bias without retraining and preserving the zero-shot generalization capabilities.

- **Future Work:** expand testing of the inference-time ablation to full-body (non-facial) datasets to assess subspace robustness.

References

[1] Ibrahim Alabdulmohsin, et al. *CLIP the Bias: How Useful is Balancing Data in Multimodal Learning?* ICLR 2024.

[2] M. D'Incà, et al. *OpenBias: Open-set Bias Detection in Text-to-Image Generative Models.* CVPR 2024.

[3] Neale Ratzlaff, et al. *Debiasing Large Vision-Language Models by Ablating Protected Attribute Representations*. NeurIPS workshop on SafeGenAl 2024.

[4] Md Mehrab Tanjim, et al. *Discovering and Mitigating Biases in CLIP-based Image Editing*. WACV 2024

[5] Tolga Bolukbasi, et al. *Man is to computer programmer as woman is to homemaker? Debiasing word embeddings*. Advances in neural information processing systems, 29, 2016

[6] Thomas Manzini, et al. *Black is to Criminal as Caucasian is to Police: Detecting and Removing Multiclass Bias in Word Embeddings*. Association for Computational Linguistics, 2019.

[7] Alec Radford, et al. *Learning Transferable Visual Models From Natural Language Supervision.* ICML 2021.

[8] https://huggingface.co/datasets/NYUAD-ComNets/

[9] Nouar AlDahoul, et al. *AI-generated faces influence gender stereotypes and racial homogenization*. 2024

[10] Boyu Han, et al. *LightFair: Towards an Efficient Alternative for Fair T2I Diffusion via Debiasing Pre-trained Text Encoders*. NeurIPS 2025

Thank you for the attention!

Claudia Cornacchia
cornacchia.1986434@studenti.uniroma1.it